

SPECIAL RELATIVITY

Invariance and Lorentz Vectors

From orthogonal transformations to Minkowski space

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1 The invariant interval

The spacetime interval

$$ds^2 = dx^2 + dy^2 + dz^2 - c^2 dt^2 \quad (1)$$

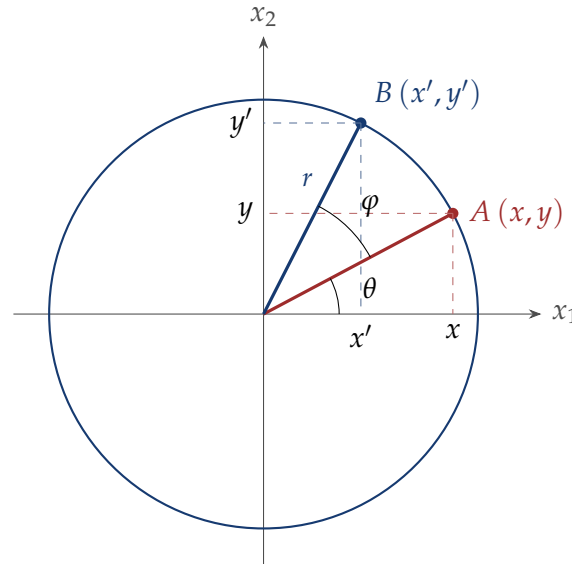
is *invariant under Lorentz transformations*.

Sign convention. Many texts write the interval with the opposite overall sign, $c^2 dt^2 - dx^2 - dy^2 - dz^2$. Either choice is invariant; these notes use the form above.

2 Orthogonal transformations

2.1 Warm-up: rotations in the plane

Under a rotation, the length of the radius is invariant. A rotation through an angle φ carries the point $A = (x, y)$ to the point $B = (x', y')$ on the same circle.



Mathematically,

$$\begin{pmatrix} x' \\ y' \end{pmatrix} = \underbrace{\begin{pmatrix} \cos \varphi & -\sin \varphi \\ \sin \varphi & \cos \varphi \end{pmatrix}}_O \begin{pmatrix} x \\ y \end{pmatrix}. \quad (2)$$

$$\sqrt{x^2 + y^2} = \sqrt{(x')^2 + (y')^2} = r, \quad O^T O = I.$$

This is called an **orthogonal transformation**. In index notation, with $x_1 = x$ and $x_2 = y$,

$$x'_i = O_{ij} x_j = \sum_j O_{ij} x_j \quad (3)$$

so that, for instance,

$$x'_1 = O_{11}x_1 + O_{12}x_2 = x \cos \varphi - y \sin \varphi.$$

2.2 Getting into 3D

With $x_1 = x$, $x_2 = y$, $x_3 = z$, the same idea reads

$$x'_i = \sum_j R_{ij} x_j \quad \text{such that} \quad R^T R = I, \quad (4)$$

again an **orthogonal transformation**.

The spacetime interval is unchanged when we go to the primed frame:

$$\begin{aligned} ds^2 &= dx^2 + dy^2 + dz^2 - c^2 dt^2 \\ &= dx'^2 + dy'^2 + dz'^2 - c^2 dt'^2 = ds'^2. \end{aligned} \quad (5)$$

Now define

$$x_1 = x, \quad x_2 = y, \quad x_3 = z, \quad x_0 = ict \quad (i = \sqrt{-1}),$$

so that $dx_0^2 = -c^2 dt^2$ and

$$ds^2 = dx_1^2 + dx_2^2 + dx_3^2 + dx_0^2. \quad (6)$$

The interval now looks like a squared length in a four-dimensional space with coordinates $(x_0, x_1, x_2, x_3) = (ict, x, y, z)$, called **Minkowski space**.

3 Lorentz vectors

A **Lorentz vector** is an object A_μ with $\mu = 0, 1, 2, 3$, where

- $\mu = 0$ is the *temporal* (time) component, and
- $\mu = 1, 2, 3$ are the *spatial* components.

Its square is a **Lorentz scalar**, invariant under Lorentz transformations (the analogue of $\vec{A} \cdot \vec{A}$ in 3D):

$$A^2 = A_\mu A^\mu = A_0 A^0 + A_1 A^1 + A_2 A^2 + A_3 A^3. \quad (7)$$

Minkowski space: $x_1 = x, \quad x_2 = y, \quad x_3 = z, \quad x_0 = ict.$

4 Raising and lowering indices

4.1 Prescription

Going between A_μ (lower index) and A^μ (upper index) uses one of two equivalent sign conventions:

$$\underbrace{\begin{cases} A_i \rightarrow A^i \\ A_0 \rightarrow -A^0 \end{cases}}_{\text{signature } (-,+,+,+)} \quad \underbrace{\begin{cases} A_i \rightarrow -A^i \\ A_0 \rightarrow A^0 \end{cases}}_{\text{signature } (+,-,-,-)} \quad (8)$$

Convention. From here on, $x^0 = ct$ denotes the *real* time coordinate; the signs above (rather than a factor of i) now carry the minus sign of the interval.

Applying the prescription to the position vector gives

$$\begin{aligned} x^2 &= x^\mu x_\mu = x^0 x_0 + x^1 x_1 + x^2 x_2 + x^3 x_3 \\ &= -(x^0)^2 + (x^1)^2 + (x^2)^2 + (x^3)^2 && \text{for } (-, +, +, +), \end{aligned} \quad (9)$$

$$= (x^0)^2 - (x^1)^2 - (x^2)^2 - (x^3)^2 \quad \text{for } (+, -, -, -). \quad (10)$$

4.2 Lorentz scalars from Lorentz vectors

If A_μ and B_μ are Lorentz vectors, then

$$A_\mu A^\mu, \quad B_\mu B^\mu, \quad A^\mu B_\mu \quad (11)$$

are all Lorentz scalars.

5 Generalised Lorentz transformation

For a boost with velocity $\vec{v}$ in an arbitrary direction, split the position vector into parts parallel and perpendicular to $\vec{v}$:

$$\vec{x} = \vec{x}_\parallel + \vec{x}_\perp, \quad \vec{x}_\parallel \parallel \vec{v}, \quad \vec{x}_\perp \perp \vec{v}. \quad (12)$$

Exercise

Find the transformation matrix in this case, given

$$\vec{x}' = \vec{x}'_\parallel + \vec{x}'_\perp,$$

$$\vec{x}'_\perp = \vec{x}_\perp,$$

$$\vec{x}'_\parallel = \gamma (\vec{x}_\parallel - \vec{v} t),$$

that is,

$$\vec{x}' = \gamma \vec{x}_\parallel - \gamma \vec{v} t + \vec{x}_\perp.$$

The worked solution appears in the next set of notes (SR_5).